

MING-CHIEH HSU

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EDUCATION

National Yang Ming Chiao Tung University (NYCU)

09/2021 – 06/2025

Bachelor of Science, Double Major: Materials Science and Engineering & Management Science

Hsinchu City, Taiwan

GPA: 4.01 / 4.3

WORK EXPERIENCE

Microcosm Technology

01/2026 – 05/2026

Process Integration Engineer (Full-time)

Tainan City, Taiwan

- Operated and maintained semiconductor photolithography equipment (coating, exposure, development), ensuring process safety, stability, and reliability throughout manufacturing.
- Introduced and qualified new materials, troubleshoot process issues, and drove optimization initiatives that improved throughput and yield while reducing manufacturing costs.

Industrial Technology Research Institute

03/2025 – 08/2025

Research Assistant (Manufacturing Intern) (Part-Time), Materials and Chemical Research Laboratories *Hsinchu City, Taiwan*
(20 hrs./wk. for a total of 25 wks.)

- Utilized measuring tools to collect and analyze production data to ensure production efficiency and high performance.
- Assisted factories with production procedures and other administrative matters including document organization and communication across departments.

RESEARCH & LAB EXPERIENCES

Dept. of Materials Science and Engineering, NYCU

02/2024 – 09/2024

Lab Assistant (Advisor: Prof. Yu-Chuan Lin)

Hsinchu City, Taiwan

(20 hrs./wk. for a total of 25 wks.)

- Supported laboratory tasks and assisted in measuring data related to experiments.
- Learned Fiji (measurement software) and organized Fiji training courses.

Advanced Materials Labs (II), NYCU

02/2025 – 06/2025

Student (Advisor: Prof. Wen-Wei Wu)

Hsinchu City, Taiwan

- Fabricated p-n diodes and semiconductor devices through end-to-end cleanroom processes, encompassing wafer cleaning, oxidation, diffusion, photolithography, etching, and metallization.
- Analyzed semiconductor device physics (including TLM) to evaluate electrical performance, while operating vacuum systems and thin-film deposition equipment.

Advanced Materials Labs (I), NYCU

09/2024 – 01/2025

Student (Advisors: Prof. Hung-Cheu Lin, Prof. Chun-Hua Chen)

Hsinchu City, Taiwan

- Synthesized polymers (PMMA, PS, Epoxy) and fabricated optical films, PDLCs, and LEDs, characterizing properties via FTIR, DSC, GPC, and EL/PL spectroscopy.
- Processed BaTiO₃ electroceramics and synthesized TiO₂ nanoparticles via hydrothermal methods, conducting structural (XRD), electrical, and UV-Vis photocatalytic characterizations.

Elementary Materials Labs (II), NYCU

02/2024 – 06/2024

Student (Advisor: Prof. E-Wen Huang)

Hsinchu City, Taiwan

- Investigated microstructures and mechanical properties of pure copper and aluminum alloys through cold working, annealing, and precipitation hardening.
- Evaluated electrochemical principles (Pourbaix diagrams) and analyzed Zn-Cu diffusion kinetics using the Kirkendall Effect, Fick's 2nd Law, and metallographic analysis.

Elementary Materials Labs (I), NYCU

09/2023 – 01/2024

Student (Advisor: Prof. Tsai-Fu Chung)

Hsinchu City, Taiwan

- Performed metallographic preparation and microstructural observation to analyze heat treatment effects on metallic materials.
- Conducted material impact and tensile testing for comprehensive mechanical property evaluation.